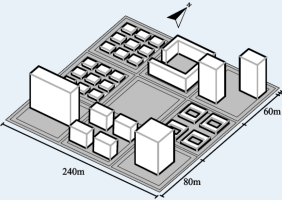


Part I - Main manuscript candidates

Final manual Fig. 1-3 PDFs and locked candidate Main Fig. 4-10.

1. Feasible morphology generation



Feasible block geometry → 12 morphology descriptors

Randomly sampled

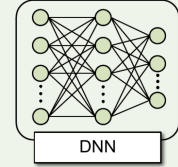
- open-space position
- building prototype
- number of floors
- block orientation θ

2. Analytic response and surrogate modeling

12 morphology descriptors

Analytic response generator

EUI / EG / H



DNN

Nested sample pools

- 500 blocks
- 1000 blocks
- 1500 blocks
- 2000 blocks

Selected DNN surrogate

Inputs: 12 descriptors
Outputs: EUI, EG, H

4. Feasible design assessment

Descriptor-space candidate

Nearest generated feasible block

Mapping distance

Mapping outcomes

- unique feasible blocks
- many-to-one candidate mapping

Cross-model assessment

- 18 directly sampled blocks
- 6 optimizer-linked blocks
- EUI / EG proxy / H

Climate sensitivity

- 4 representative blocks
- Dongtai baseline
- Beijing / Guangzhou / Harbin

3. Surrogate-assisted optimization

Shared surrogate environment

Selected DNN + guardrail + output clipping

DDPG

- 20 seeds per setting
- 600 episodes per seed
- 40 queries per episode

NSGA-II

- 20 independent runs
- 24,000 queries per run

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}$$

$$u = (0, 1, 1)$$

$$d_w = \frac{\|w \odot (z - u)\|_2}{\|w\|_2}, \quad R = 1 - d_w$$

w : axis-scaling coefficients

Additional comparison

- CMA-ES
- RandomSearch

candidate solutions

selected surrogate

Manual Fig. 1 candidate (Fig1 manual_fig1_candidate)

Source files: paper/manuscript/figures/round2_candidate/manual/fig1.pdf

Source SHA: ba37052f7cdb...

Claim boundary: Manual Fig. 1 is included for visual QA and gallery context only; the automated workflow must not edit or re-export it.

Revision note: Final user-supplied manual Fig. 1 PDF; included for gallery and QA only, with no automatic edits.

Unresolved concerns: None

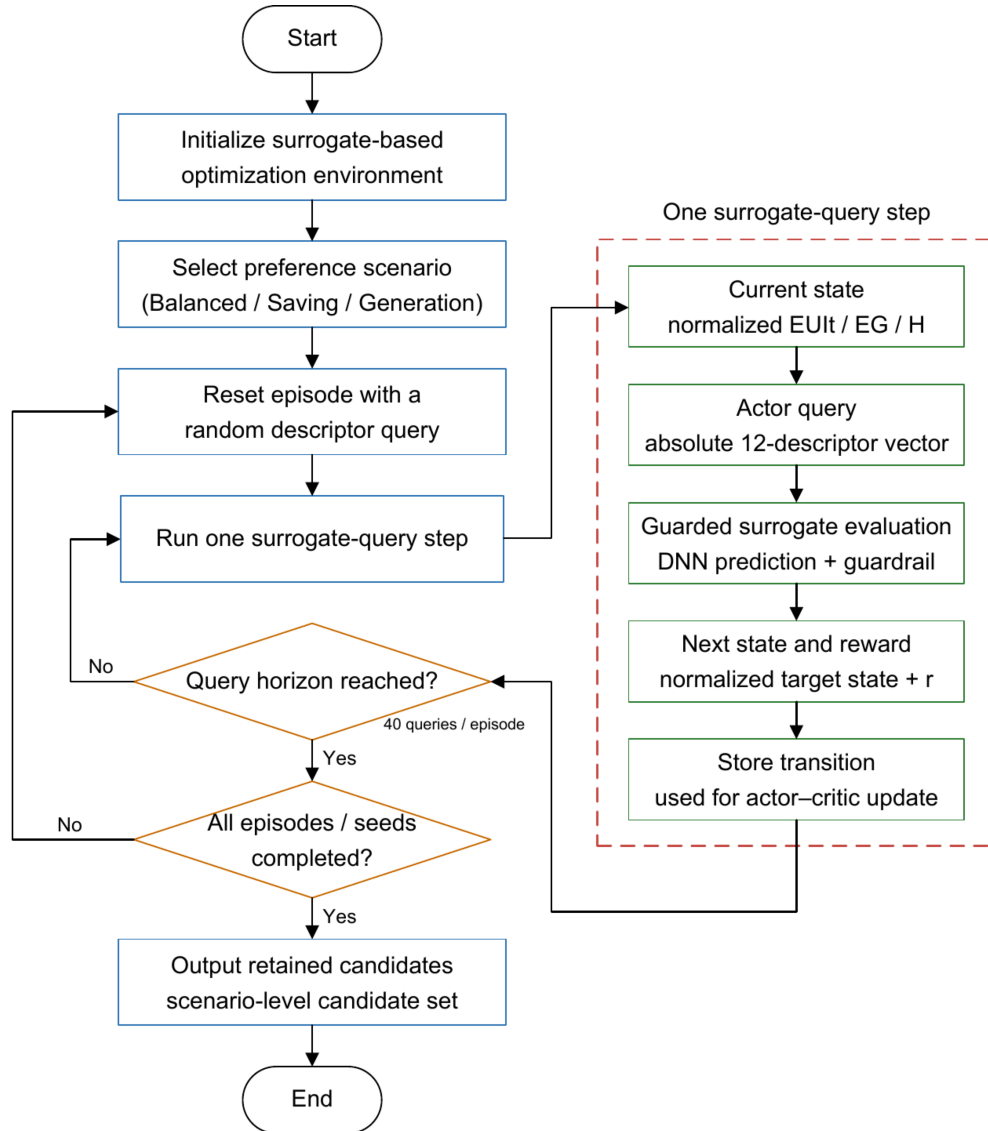


Fig. 2 manual workflow candidate (Fig2 manual_fig2_workflow)
Source files: paper/manuscript/figures/round2_candidate/manual/fig2.pdf
Source SHA: c97bdb378e04...
Claim boundary: The figure describes the DDPG-based surrogate-search workflow and single-query surrogate interaction; actor-critic learning mechanics are reserved for Fig. 3.
Revision note: Final user-supplied manual Fig. 2 workflow PDF; TeX round2/round3 candidates are no longer preferred.
Unresolved concerns: None

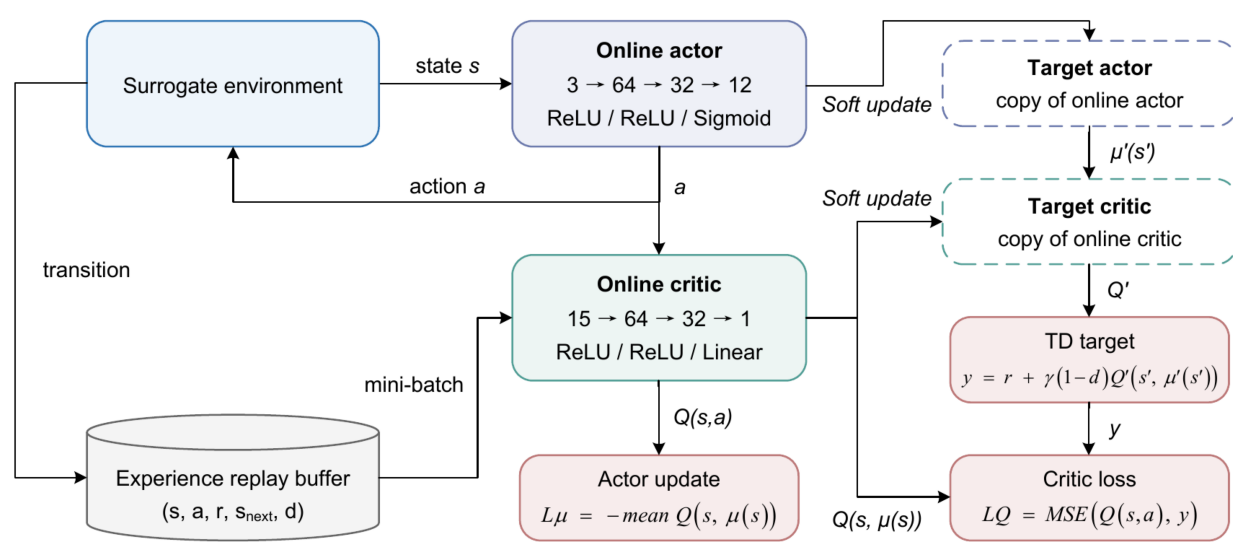


Fig. 3 manual DDPG architecture candidate (Fig3 manual_fig3_ddpg_architecture)

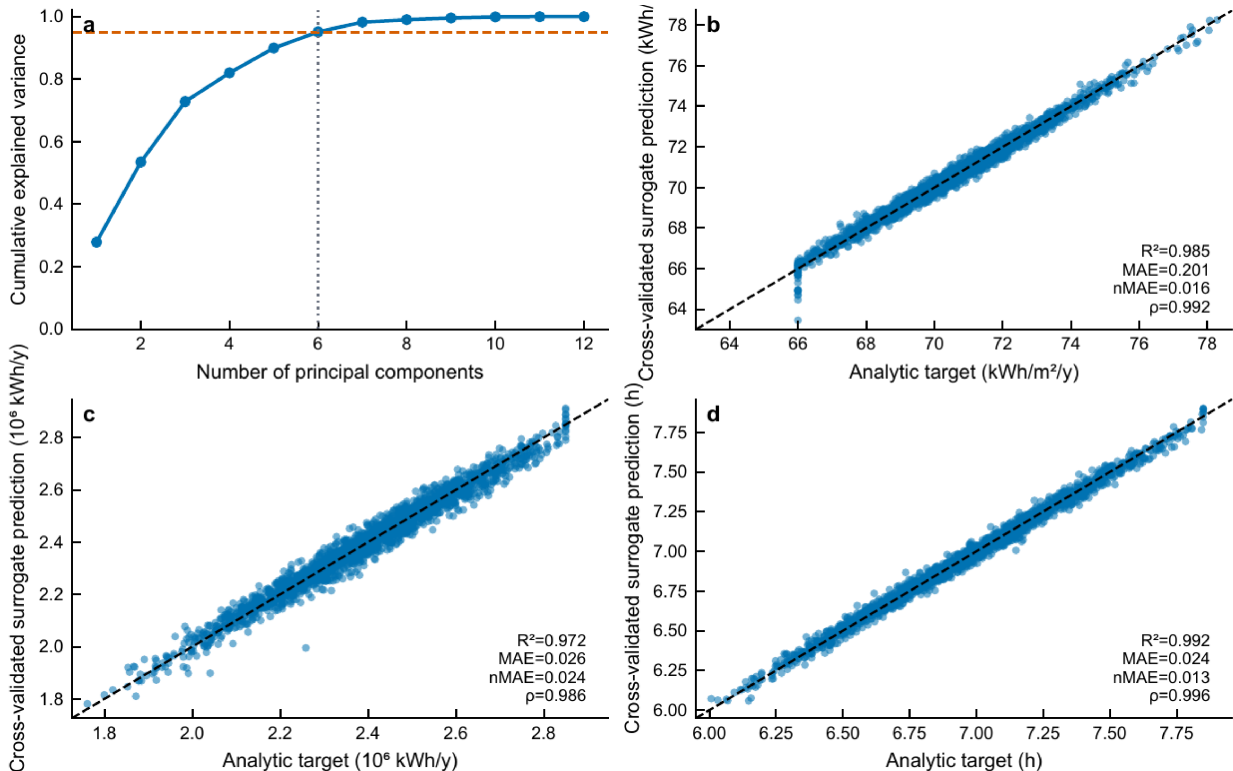
Source files: paper/manuscript/figures/round2_candidate/manual/fig3.pdf

Source SHA: 83b9f65099de...

Claim boundary: The figure documents the DDPG learning architecture used in surrogate-assisted optimization and does not describe workflow termination or episode sequencing.

Revision note: Final user-supplied manual Fig. 3 DDPG architecture PDF; TeX round2/round3 candidates are no longer preferred.

Unresolved concerns: None



Main Fig. 4 (M1 data_and_surrogate_validation)

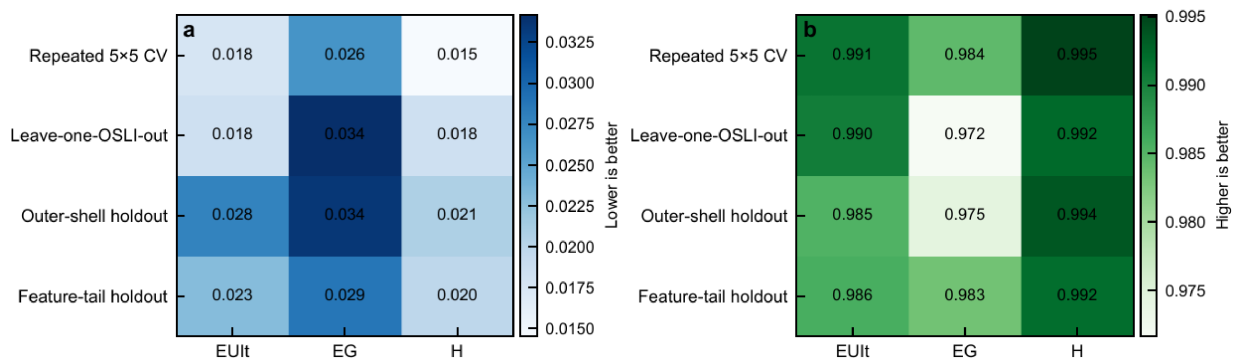
Source files: descriptor_coverage.csv, surrogate_parity_mean_predictions.csv

Source SHA: cefa7377e3fb..., 5837b986d33d...

Claim boundary: Supports descriptor-space coverage and analytic-target surrogate fidelity only.

Revision note: Carried forward from the canonical round-2 candidate set without a layout change in this task.

Unresolved concerns: None



Main Fig. 5 (M2 surrogate_robustness)

Source files: surrogate_validation_regimes.csv

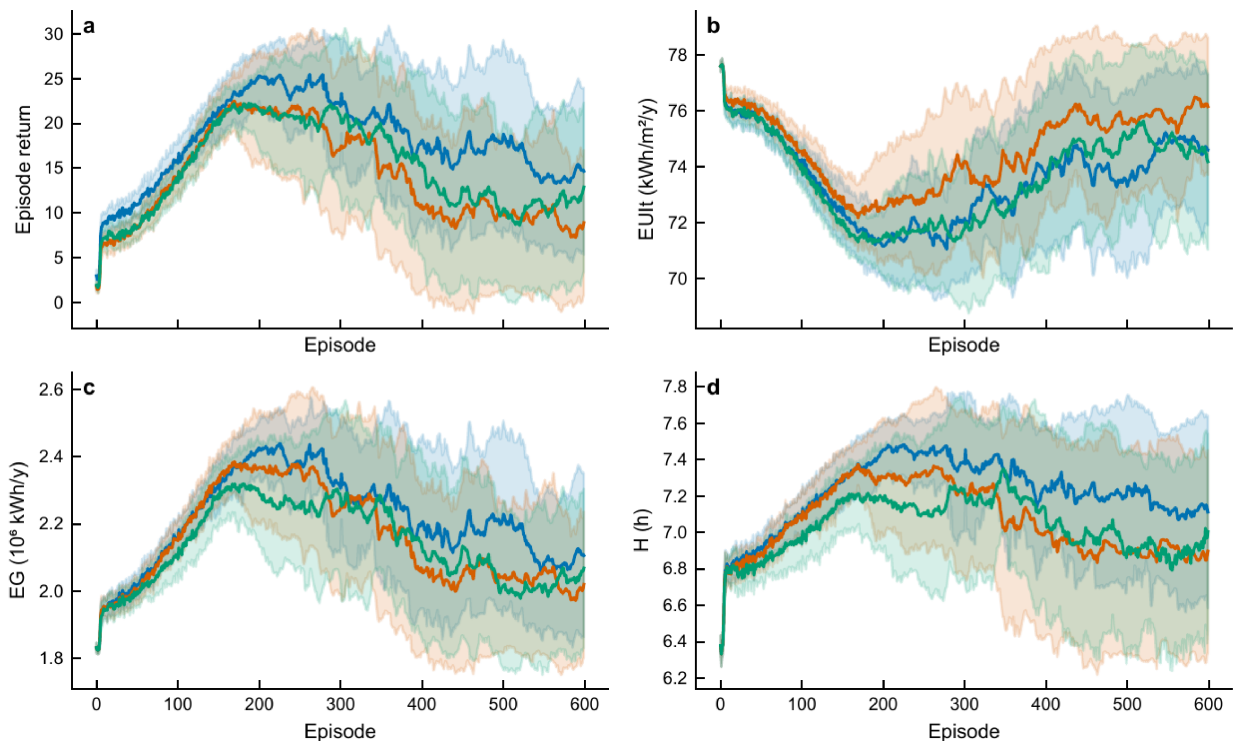
Source SHA: 4a35f8c39487...

Claim boundary: All panels remain analytic-target surrogate-validation evidence, not physical validation.

Revision note: Carried forward from the canonical round-2 candidate set without a layout change in this task.

Unresolved concerns: None

—●— Balanced Performance —▲— Energy Saving Focus —■— Energy Generation Focus



Main Fig. 6 (M3 ddpg_training_dynamics)

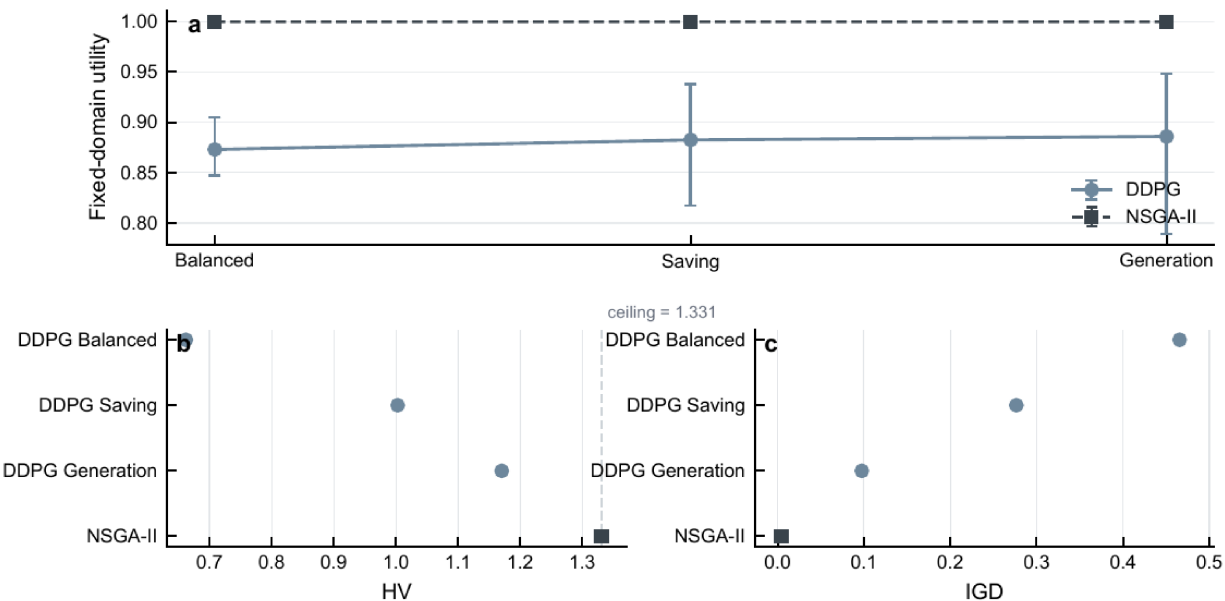
Source files: ddpg_training_curves_summary.csv, ddpg_seed_diagnostics.csv

Source SHA: 36a74af74f74..., bf356c2b9859...

Claim boundary: Training dynamics describe serialized surrogate-query search only.

Revision note: Carried forward from the canonical round-2 candidate set without a layout change in this task.

Unresolved concerns: None



Main Fig. 7 (M4 benchmark_fairness)

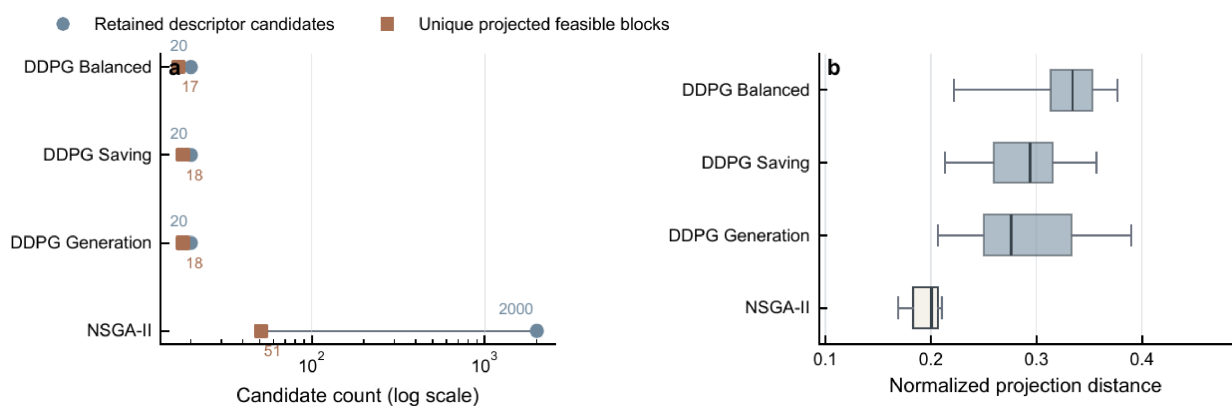
Source files: benchmark_utility.csv, benchmark_equal_size_20.csv, benchmark_output_contract_counts.csv

Source SHA: c8dafd40dd57..., f60b07256e7b..., 83c911df9c51...

Claim boundary: Equal-size metrics remain benchmark-reference-v2 evidence only; broader contract and diagnostic baselines are deferred to Supplementary Information.

Revision note: Simplified the main benchmark figure to fixed-domain utility plus matched-size HV/IGD for DDPG and NSGA-II only.

Unresolved concerns: None



Main Fig. 8 (M5 feasible_projection)

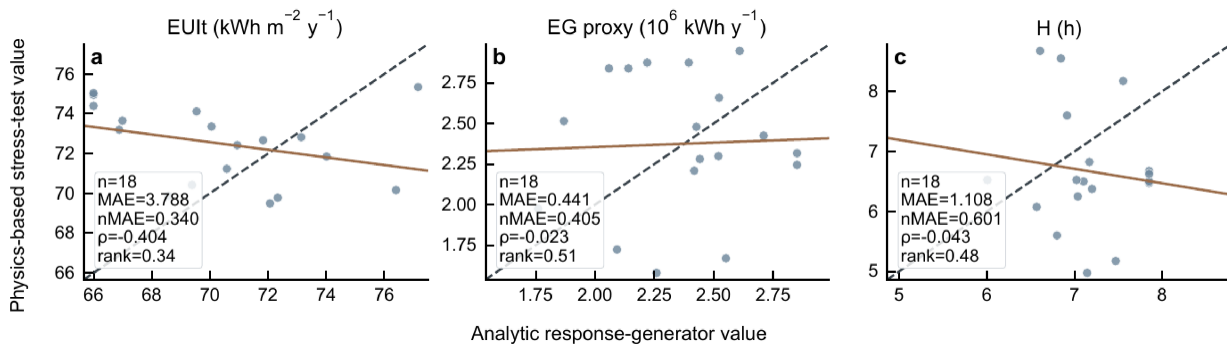
Source files: feasible_projection_summary.csv, feasible_projection_metrics.csv

Source SHA: 16f00a0bdf86..., 1132a919d3db...

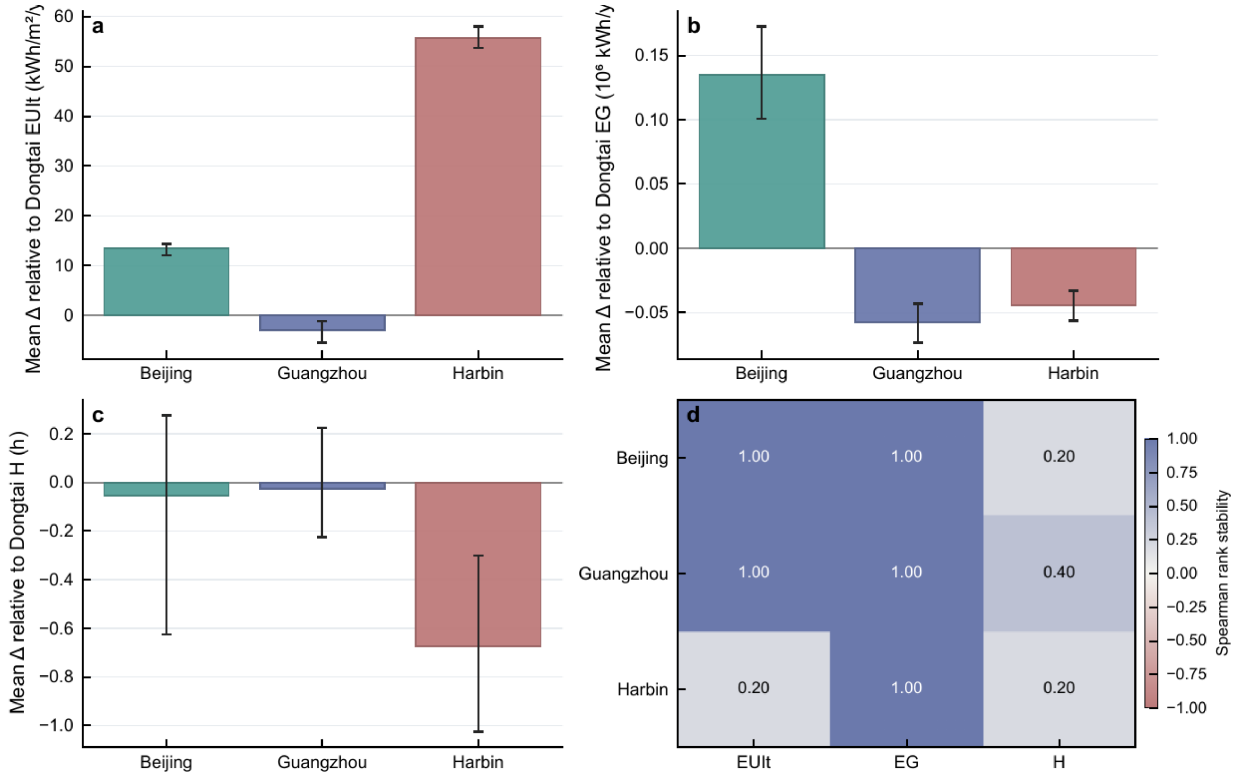
Claim boundary: Projection remains a nearest-neighbour representation diagnostic rather than physical validation.

Revision note: Reduced the projection figure to representation compression plus projection-distance diagnostics for the main-text comparison.

Unresolved concerns: None



Main Fig. 9 (M6 physical_cross_model_stress_test)
Source files: physical_direct_cases.csv, physical_stress_metrics.csv
Source SHA: 0403f94ea9bd..., cf9ec8e791d5...
Claim boundary: This figure is limited to the direct-case physics-based cross-model stress test and does not support optimizer-superiority claims.
Revision note: Kept the same 18 direct cases while tightening axis wording, typography, marker size, and statistics placement.
Unresolved concerns: None



Main Fig. 10 (M7 cross_climate_sensitivity)

Source files: climate_case_results.csv, climate_summary.csv, climate_rank_stability.csv

Source SHA: f624253730cd..., 7a03bcf3714f..., ebd3f1d2ec54...

Claim boundary: This figure is a limited four-block cross-climate physical sensitivity analysis only.

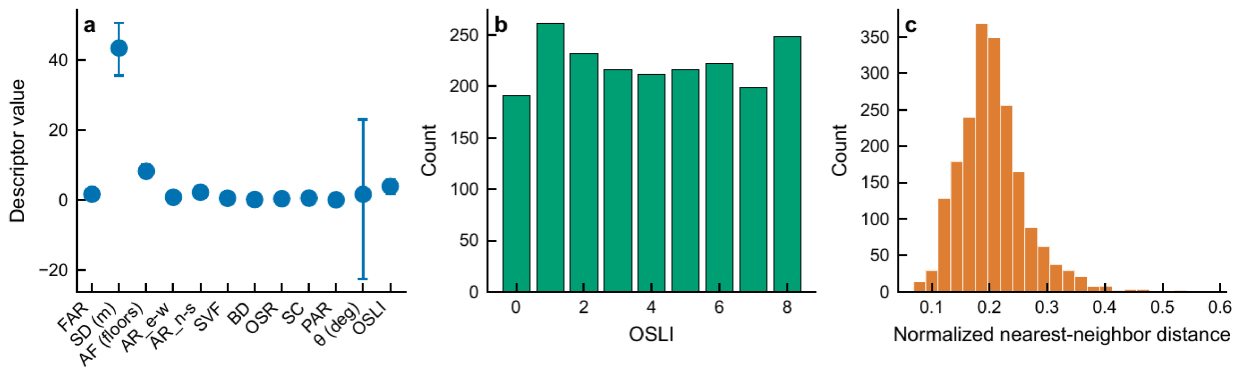
Revision note: Kept the same climate data while switching to the muted climate palette and a zero-centered low-saturation heatmap.

Palette note: Beijing #539F97, Guangzhou #6C7AAD, Harbin #BE7A7A

Unresolved concerns: None

Part II - Supplementary Information candidates

Locked candidate figures for Supplementary Fig. S1-S9.



Supplementary Fig. S1 (S1 A1_descriptor_distributions)

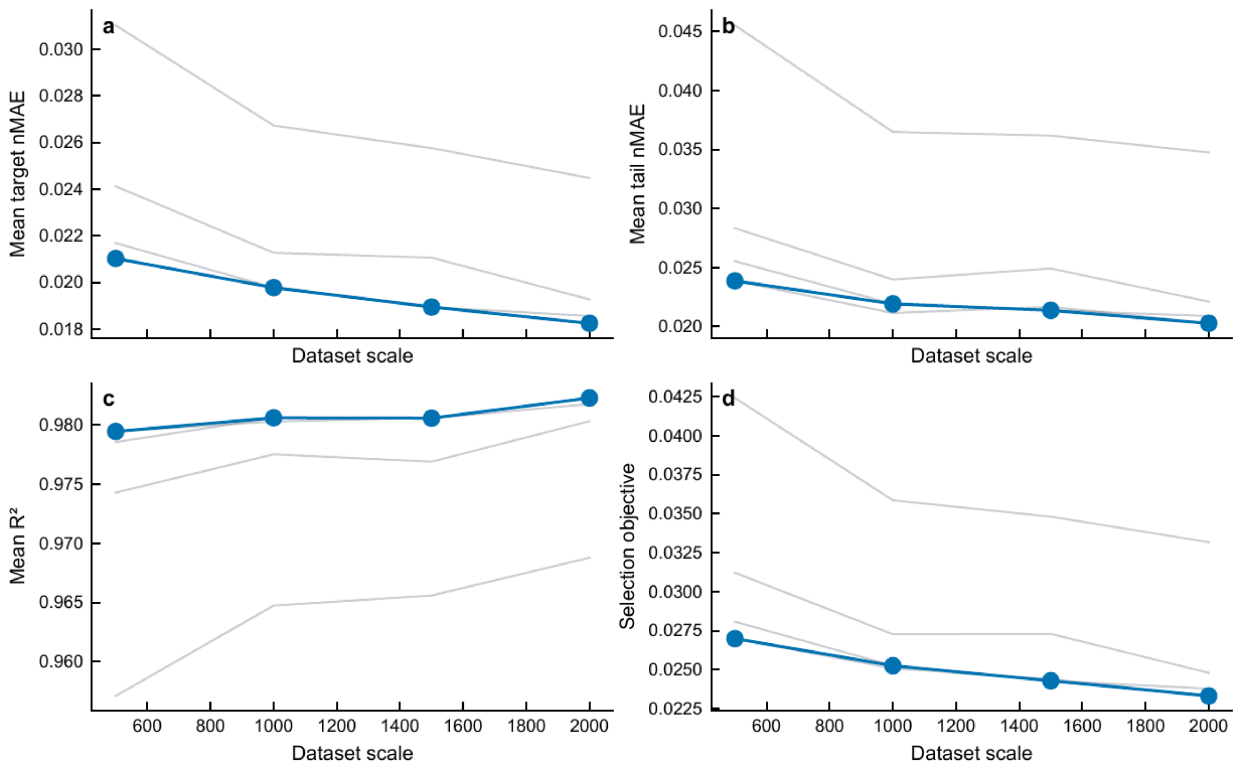
Source files: descriptor_coverage.csv

Source SHA: cefa7377e3fb...

Claim boundary: Descriptive coverage diagnostics only.

Revision note: Carried forward from the canonical round-2 candidate set and relabeled for the Supplementary Information split.

Unresolved concerns: None



Supplementary Fig. S2 (S2 A3_scale_study)

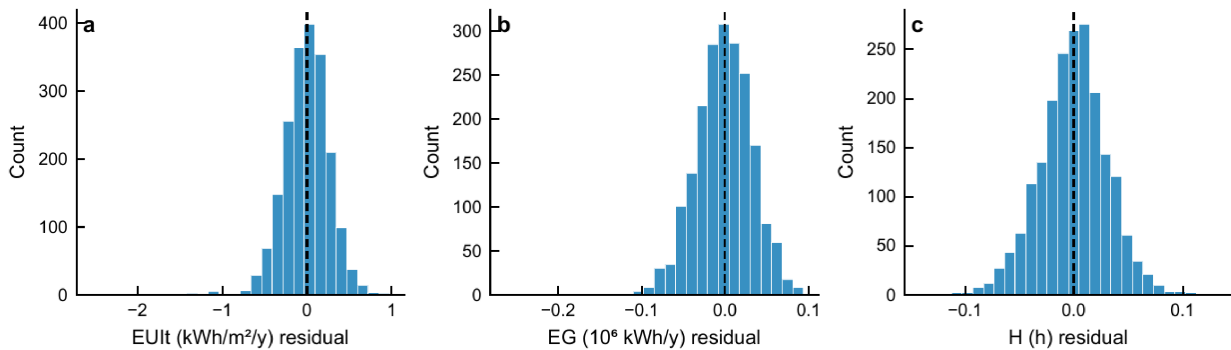
Source files: scale_study.csv

Source SHA: 87120cdcdb9...

Claim boundary: Scale-study rows support the surrogate-selection rationale only.

Revision note: Carried forward from the canonical round-2 candidate set and relabeled for the Supplementary Information split.

Unresolved concerns: None



Supplementary Fig. S3 (S3 A2_residual_diagnostics)

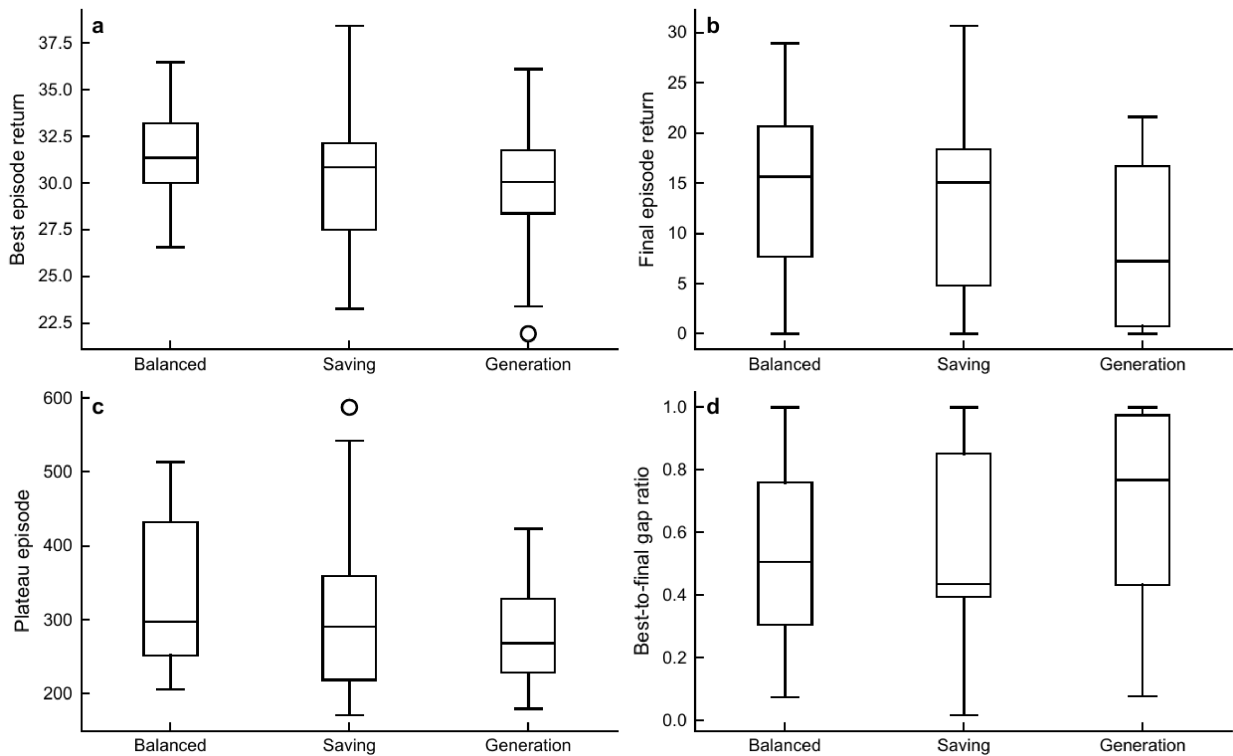
Source files: surrogate_parity_mean_predictions.csv

Source SHA: 5837b986d33d...

Claim boundary: Residual diagnostics describe analytic-target surrogate error only.

Revision note: Carried forward from the canonical round-2 candidate set and relabeled for the Supplementary Information split.

Unresolved concerns: None



Supplementary Fig. S4 (S4 B1_seed_diagnostics)

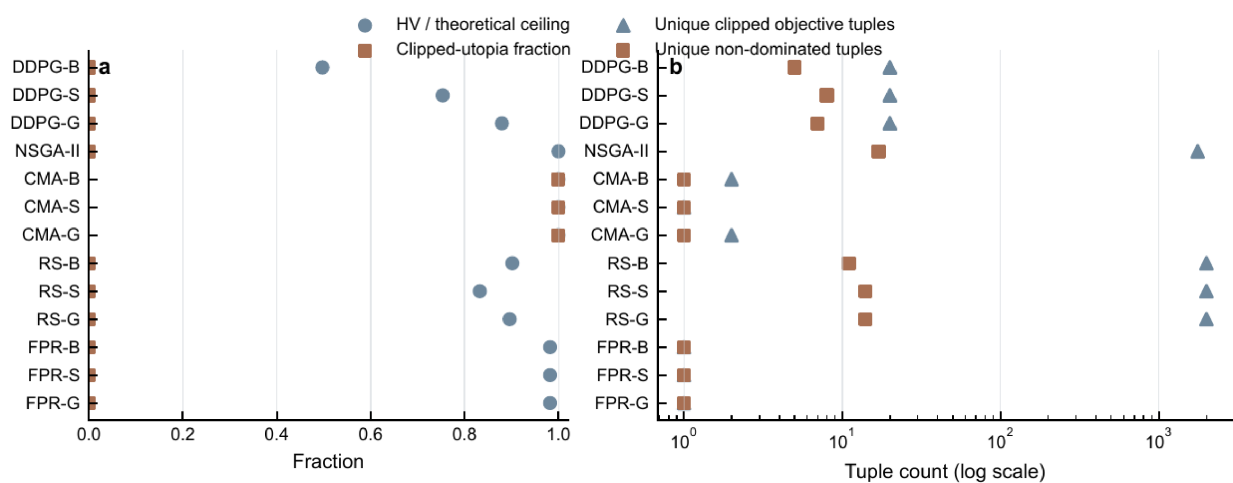
Source files: [ddpg_seed_diagnostics.csv](#)

Source SHA: [bf356c2b9859...](#)

Claim boundary: Seed diagnostics remain Supplementary Information training evidence only.

Revision note: Carried forward from the canonical round-2 candidate set and relabeled for the Supplementary Information split.

Unresolved concerns: None



Supplementary Fig. S5 (S5 B3_hv_ceiling_diagnostics)

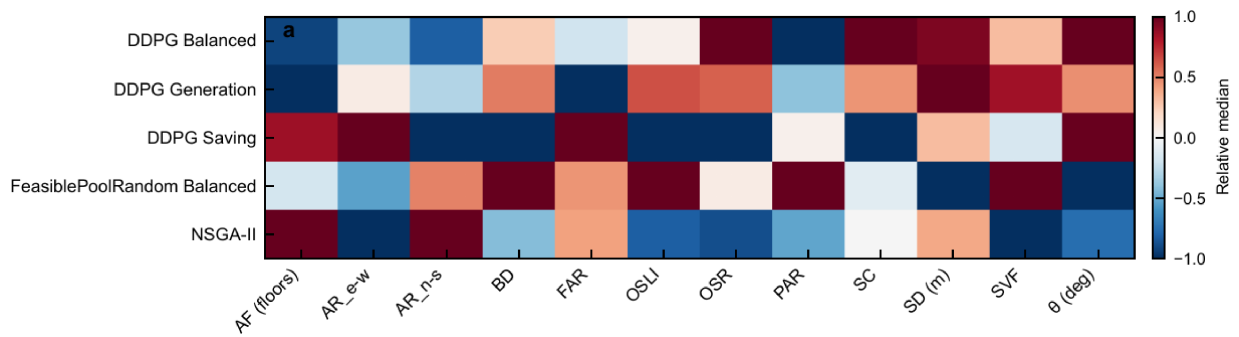
Source files: benchmark_hv_ceiling.csv

Source SHA: fa862502f196...

Claim boundary: HV ceiling and tuple-collapse panels remain supplementary diagnostics only.

Revision note: Moved ceiling and duplicate-collapse diagnostics into a readable two-panel Supplementary Information layout with short labels.

Unresolved concerns: None



Supplementary Fig. S6 (S6 B2_morphology_signatures)

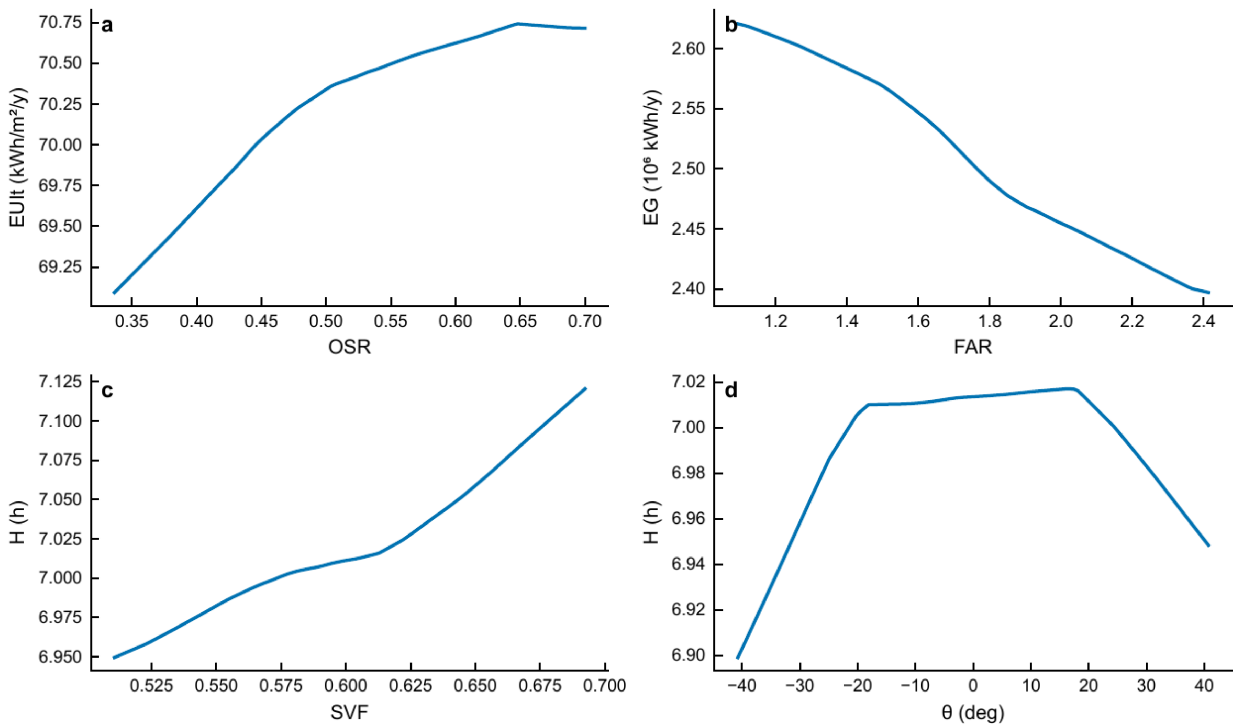
Source files: morphology_signatures.csv

Source SHA: aa6df0731235...

Claim boundary: Descriptor signatures are descriptive summaries, not stable design rules.

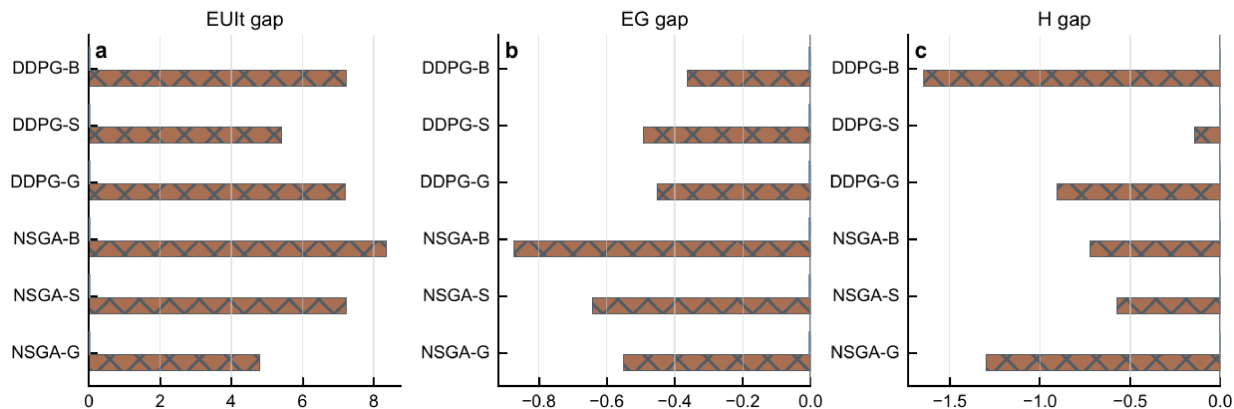
Revision note: Carried forward from the canonical round-2 candidate set and relabeled for the Supplementary Information split.

Unresolved concerns: None



Supplementary Fig. S7 (S7 B5_nonlinear_response_profiles)
Source files: scale_study.csv
Source SHA: 87120cdcdb9...
Claim boundary: Selected surrogate response profiles illustrate local trends only.
Revision note: Carried forward from the canonical round-2 candidate set and relabeled for the Supplementary Information split.
Unresolved concerns: None

Projection gap Cross-model gap



Supplementary Fig. S8 (S8 B4_optimizer_linked_gap_decomposition)

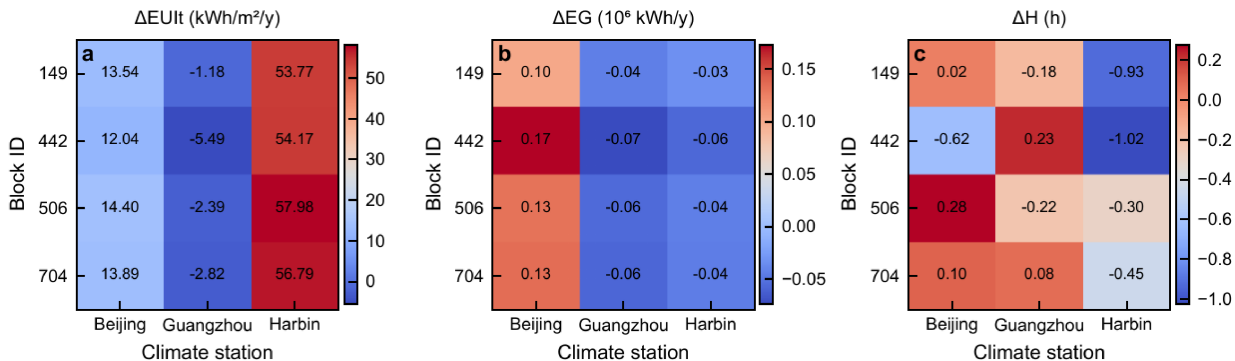
Source files: optimizer_linked_physical_gaps.csv

Source SHA: 79d6291c8305...

Claim boundary: These optimizer-linked cases are representative bridge diagnostics rather than a global optimizer benchmark.

Revision note: Rebuilt the optimizer-linked bridge diagnostics as horizontal gap bars with an out-of-panel legend and short case labels.

Unresolved concerns: None



Supplementary Fig. S9 (S9 B6_climate_case_detail)
Source files: climate_case_results.csv
Source SHA: f624253730cd...
Claim boundary: Case-level climate details remain limited to four blocks and three additional climates.
Revision note: Carried forward from the canonical round-2 candidate set and relabeled for the Supplementary Information split.
Unresolved concerns: None